

TFLite Converter segfaults when trying to convert per-channel quantized transposed convolutions #53767

✓ Closed



lgeiger opened on Jan 15, 2022 · edited by lgeiger

Edits ▾ Contributor ...

When converting transposed convolutions using per-channel weight quantization the converter segfaults and crashes the Python process. Per-channel quantization is supported by TFLite Transposed convolutions:

[tensorflow/tensorflow/lite/kernels/transpose_conv.cc](https://github.com/tensorflow/tensorflow/blob/master/tensorflow/lite/kernels/transpose_conv.cc)

Lines 371 to 380 in f87be6c

```
371     TF_LITE_ENSURE_EQ(context, weights->quantization.type,
372                        kTfLiteAffineQuantization);
373     const auto* affine_quantization =
374         reinterpret_cast<TfLiteAffineQuantization*>(
375             weights->quantization.params);
376     const int channels_out = weights->dims->data[0];
377     TF_LITE_ENSURE(context, affine_quantization);
378     TF_LITE_ENSURE(context, affine_quantization->scale);
379     TF_LITE_ENSURE(context, (affine_quantization->scale->size == 1 ||
380                             affine_quantization->scale->size == channels_out));
```

so the converter shouldn't segfault when trying to convert such a model.

It looks like this issue has been introduced in TensorFlow 2.6 since the same model code produced a valid TFLite file in TensorFlow 2.5. This issue might also be related to [#53766](#), but in any case the converter should never segfault.

1. System information

- OS Platform and Distribution (e.g., Linux Ubuntu 16.04): macOS / Ubuntu
- TensorFlow installation (pip package or built from source): pip package
- TensorFlow library (version, if pip package or github SHA, if built from source): 2.6, 2.7, 2.8rc0 and 2.9.0-dev20220114

2. Code

A minimal reproduction of the issue and a workaround is available in [this notebook](#).

```
import tensorflow as tf
```



```

class QuantConv2DTransposed(tf.keras.layers.Layer):
    def build(self, input_shape):
        self.kernel = self.add_weight("kernel", [3, 3, input_shape[-1], 24])

    def call(self, inputs):
        filters = tf.quantization.fake_quant_with_min_max_vars_per_channel(
            self.kernel, -3.0 * tf.ones([24]), 3.0 * tf.ones([24]), narrow_range=True
        )
        filters = tf.transpose(filters, (0, 1, 3, 2))
        return tf.nn.conv2d_transpose(inputs, filters, [*inputs.shape[:-1], 24], 1)

inp = tf.keras.Input(shape=(6, 8, 48), batch_size=1)
x = tf.quantization.fake_quant_with_min_max_vars(inp, -3.0, 3.0, narrow_range=True)
x = QuantConv2DTransposed()(x)
x = tf.quantization.fake_quant_with_min_max_vars(x, -3.0, 3.0, narrow_range=True)

model = tf.keras.Model(inp, x)

model.save("/tmp/testing")
converter = tf.lite.TFLiteConverter.from_saved_model("/tmp/testing")
converter.optimizations = [tf.lite.Optimize.DEFAULT]

# terminated by signal SIGSEGV (Address boundary error)
tflite_model = converter.convert()

```

1



 **lgeiger** added **TFLiteConverter** on Jan 15, 2022



 **google-ml-butler** assigned sushreebarsa on Jan 15, 2022



 **tilakrayal** mentioned this on Jan 17, 2022

✓ Constant folding fails when converting int8 transposed convolutions #53766



 **sushreebarsa** added **TF 2.7** **type:bug** on Jan 17, 2022

 **sushreebarsa** on Jan 17, 2022

Contributor ...

@Saduf2019 Was able to replicate the issue on colab using TF v[2.6.0](#), [2.7.0](#) and [tf-nightly](#)(2.9.0-dev20220114), please find the attached gists. Thank you!



 **sushreebarsa** assigned Saduf2019 and unassigned sushreebarsa on Jan 17, 2022

 **lgeiger** on Jan 17, 2022

Contributor Author ...

Was able to replicate the issue on colab using TF v2.6.0, 2.7.0 and tf-nightly(2.9.0-dev20220114), please find the attached gists

[@sushreebarsa](#) Thanks for confirming. Just for reference, your reproduction on TF 2.6.0 actually now fails due to an unrelated Keras version conflict. Changing the dependency from `v2.6.0` to `v2.6.2` will fix this and allow you to correctly reproduce the segfault mentioned in this issue.

1

  **Saduf2019** assigned [jvishnuvardhan](#) and unassigned [Saduf2019](#) on Jan 18, 2022

  **jvishnuvardhan** assigned [JunyoungLim](#) and unassigned [jvishnuvardhan](#) on Jan 21, 2022

  **jvishnuvardhan** added `stat:awaiting tens...` on Jan 21, 2022

 lgeiger on Mar 18, 2022

Contributor

Author

...

This is still an issue in `2.9.0-dev20220318`. Are there any updates on this?
Being able to trigger a converter segfault seems to be quite problematic.

 lgeiger on May 12, 2022

Contributor

Author

...

[@JunyoungLim](#) I retested the above example `2.10.0-dev20220427` and the converter still segfaults.

 lgeiger on Sep 13, 2022

Contributor

Author

...

I retested the above example with `2.10.0` and the segfault seems to be fixed now, however conversion still fails with:

```
'tf1.transpose' op has mismatched quantized axes of input and output
```



See [here](#).

 aniruthraj on Oct 17, 2024

Contributor

...

Hi,

Thank you for opening this issue. Since this issue has been open for a long time, the code/debug information for this issue may not be relevant with the current state of the code base.

The Tensorflow team is constantly improving the framework by fixing bugs and adding new features. We suggest you try the latest TensorFlow [version](#) with the latest compatible hardware configuration which could potentially resolve the issue. If you are still facing the issue, please create a new GitHub issue with your latest findings, with all the debugging information which could help us investigate.

Please follow the [release notes](#) to stay up to date with the latest developments which are happening in the Tensorflow space.



Venkat6871 added

stat:awaiting resp...

and removed

stat:awaiting tens...

on Oct 17, 2024



github-actions (bot) on Oct 25, 2024 – with [GitHub Actions](#)

...

This issue is stale because it has been open for 7 days with no activity. It will be closed if no further activity occurs. Thank you.



github-actions added

stale

on Oct 25, 2024



github-actions (bot) on Nov 1, 2024 – with [GitHub Actions](#)

...

This issue was closed because it has been inactive for 7 days since being marked as stale. Please reopen if you'd like to work on this further.



github-actions closed this as completed on Nov 1, 2024



google-ml-butler (bot) on Nov 1, 2024 – with [Google-ML-Butler](#)

...

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Assignees



JunyoungLim

Labels

TF 2.7

TFLiteConverter

stale

stat:awaiting response

type:bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

 Code with agent mode



No branches or pull requests

Participants



+2